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PROJECT TITLE:- AI-Powered Smart Expense Tracker with Intelligent Input, Behavioral Analysis, and Financial Chatbot

ABSTRACT

In the current digital era, managing personal finances effectively has become increasingly difficult due to unstructured spending habits and limited financial awareness. Most existing expense tracking applications focus only on recording transactions and fail to provide meaningful insights, predictive analysis, or interactive assistance. To address these limitations, this project proposes the development of an AI-powered smart expense tracking system that enhances traditional financial management. The system allows users to record expenses through both manual entry and an intelligent input feature, where unstructured text such as transaction messages can be processed to automatically extract key details like amount, category, and description. It also incorporates a virtual balance tracking mechanism in which users provide their income or budget, and the system dynamically calculates the remaining balance while generating alerts for overspending, low balance, and budget limits.

Furthermore, the application integrates behavioral analysis and prediction capabilities to identify spending patterns and estimate future expenses based on historical data, enabling users to make informed financial decisions. An interactive financial chatbot is included to improve user engagement, allowing users to ask queries in natural language and receive context-aware responses related to their expenses, balance, and savings suggestions. The system is implemented using modern technologies such as Next.js, MongoDB Atlas, Tailwind CSS, and Chart.js, ensuring scalability, responsiveness, and efficient data visualization. By combining intelligent input processing, analytical insights, and interactive assistance, the proposed system transforms a basic expense tracker into a smart financial assistant that promotes better financial planning and responsible spending.